Lab 07 — Model Evaluation

Group 4 — Reflection Journal

Course:ITAI 1371 — Introduction to Machine Learning

Lab:Module 07 — Better Model Evaluation

Group: Group 4

Member 1 — Patrick Rostand Gandjouon Tchassem

Reflection:

This lab helped me understand that accuracy alone is not enough to decide whether a machine learning model is good. In our previous lab, we focused more on training models and getting predictions. In this lab, I learned how to evaluate those predictions and understand the mistakes the model makes.

One thing that stood out to me was the confusion matrix. Our model correctly predicted 90 passengers who died and 54 who survived, but it also made 35 incorrect predictions. This helped me see why precision and recall are important. For example, our model had 73% recall for passengers who survived, meaning it missed some of the actual survivors. I now understand that the type of mistake matters depending on the problem we are trying to solve.

I also learned why cross-validation is useful. Our model had about 80% accuracy on the test data, while the average cross-validation score was 78.68%. Seeing these results helped me understand that one train-test split does not always tell the whole story. Cross-validation gives us a better idea of how consistently a model performs on different parts of the data. Overall, this lab improved my understanding of model evaluation and showed me why checking errors is just as important as training the model.

Member 2 — Saimi Manasiya.

In this lab, I learned how to evaluate a machine learning model using different evaluation techniques instead of depending only on accuracy. I worked with a Logistic Regression model using the Titanic dataset to predict whether passengers survived. I learned how to create and interpret a confusion matrix, which helps identify correct and incorrect predictions, including true positives, true negatives, false positives, and false negatives. I also learned about precision, recall, and the F1-score and understood how these metrics help us measure the performance of a classification model in different ways. Additionally, I practiced generating a classification report and performing five-fold cross-validation to check how consistently the model performs on different parts of the dataset. The average cross-validation score was 78.68%. Overall, this lab improved my understanding of model evaluation and taught me why it is important to use multiple evaluation methods to understand a machine learning model's reliability and performance.

Member 3 — Kenneth Kouokam

For Module 07, my focus is interpreting the classification report and understanding what its scores mean. The main lesson I take from this lab is that a single accuracy number can hide important differences in a model's predictions. About 80% accuracy sounds encouraging, but it does not explain which passengers the model identifies well or which mistakes it makes. Looking at each class separately gives me a more useful way to judge the results.

The difference between precision and recall becomes clearer when I connect the formulas to the group's reported confusion matrix. The model predicts survival for 69 passengers, and 54 of those predictions are correct, giving precision of about 78%. However, there are 74 actual survivors, and the model identifies only 54 of them, giving recall of about 73%. These percentages answer different questions. Precision asks how often a survival prediction is correct, while recall asks how many of the actual survivors the model finds.

The 20 missed survivors help explain why accuracy alone is not enough. Those passengers are false negatives when survival is the positive class. The 15 passengers predicted to survive who actuallydied are false positives. Both errors reduce performance, but they different weaknesses. Comparing the two classes also shows that recall is higher for passengers who died, at about 86%. This means that the overall score does not represent

equal performance for both groups, even though it summarizes all of their predictions together.

F1-score adds another useful perspective because it combines precision and recall using their harmonic mean. For survivors, the score is about 76%, which reflects both the incorrect survival predictions and the survivors the model misses. I also see why support belongs in the report: the test set contains 105 passengers who died and 74 survivors. Knowing those counts gives the percentages context and reminds me to look at how many examples each score represents.

This lab connects to Module 06 by extending the question from how accurate a classifier is to what kinds of errors it makes. It also raises a question I would want to explore: could we identify more survivors without creating too many incorrect survival predictions? The answer would require further testing rather than assuming one metric tells the whole story. My takeaway is to explain the numbers in terms of actual predictions and to consider which errors matter for the problem. A useful evaluation should make both the model's strengths and its limitations understandable.

Member 4 — Anavictoria Cavazos

This week's assignment, Lab 07, helped me understand more on how confusion matrix is read and used. After imputing the proper code I saw how 90 of the total people were predicted to survive and that was true, but then 20 were predicted to have survived when in fact they died. Now this error according to what I learned is known as a type 2 error, which at first I did not understand why or the difference between type 1 and that one. Then after looking deeper into the assignment/lesson I understood that for a type 1 error it is and error of claiming a negatve, but it is not, so in the case of the titanic dataset it is seen when there was a claim of predicting the ones that died, but they actually survived.

Going further into my reflection I understood the importance of splitting the data as mentioned in class and practiced in past assignments, but now having a hands on with seeing how splitting the data is just the first step and then there has to be cross-validation, wherethe splitting has to happen multiple times helped me see the reason for it and the importance. The way that it affects the data is by improving the accuracy by seeing the true number of how great the model is and where exactly are its flaws at. For instance, when it finally reaches its last cross validation, it is easier to see where the model did not perform well, so then that can be fixed and adjusted to perform to the best it can.

Member 5 — Hashim Sayed Hoosini

In this model evaluation lab, I transitioned from relying on a single, potentially misleading metric like accuracy to employing a standard set of robust evaluation techniques. I learned that accuracy alone cannot reveal whether a machine learning model is truly trustworthy, and it is always best practice to evaluate other metrics as well. Rather than looking at a single accuracy score, mapping the confusion matrix helped me see exactly where the model's predictions deviated. For example, knowing that we had 90 True Negatives, 54 True Positives, 15 False Positives, and 20 False Negatives provided immediate, actionable context.

Exploring the precision-recall trade-off highlighted the real-world implications of prioritizing precision (minimizing false positives, such as in spam detection) versus recall (minimizing false negatives, such as in medical diagnoses). This trade-off demonstrated that an optimal model is highly dependent on the specific business context and the unique costs associated with incorrect classifications.

Additionally, I learned to use cross-validation as a trust anchor. Performing a 5-fold cross-validation confirmed that our initial single train-test split was genuinely representative. The stable average score of 78.68% across all folds validated that the model's performance was not merely a product of random chance.

Ultimately, this lab underscored why thoroughly validating machine learning models is so crucial. In a real-world project, deploying a model based purely on a high test accuracy score can be incredibly costly if the minority class is heavily misclassified.